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CHE 321; Spring - 2020; Quiz - 4

100 Points

A reversible reaction, $A + B \leftrightarrow C + D$ ($\Delta H_{rxn} = -100 \text{ KJ/mol}$), is taking place in a CSTR at adiabatic conditions. The rate law for the reaction is:

$$-r_A = k \left(C_A C_B - \frac{1}{K} C_C C_D \right)$$

exothermic

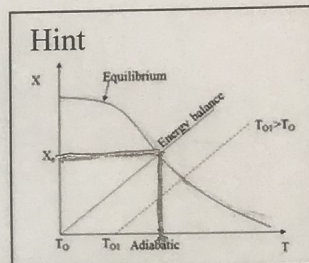
$$K_0 = 3.5673; H_1/R = 600.32 \text{ K}$$

$$K = K_0 \exp\left(\frac{H_1}{RT}\right)$$

1. If for a particular run in the CSTR, the conversion is found to be 0.8, what can you say about the maximum exit temperature possible for the CSTR? [50]

$$K = 3.5673 \exp\left(\frac{600.32 \text{ K}}{T}\right)$$

$$\ln(K) = \frac{600.32 \text{ K}}{T} \ln(3.5673)$$



2. For a CSTR with heat exchanger (typical for the reaction above) in which the above reaction is being carried out, the following equation represents the energy balance. [25]

$$\underbrace{\dot{Q}}_1 + \underbrace{[-\Delta H_{Rx}^0(T_R) - \Delta \hat{C}_p(T - T_R)]F_{A0}X}_{2} + \underbrace{(-F_{A0}) \sum_{i=1}^n \Theta_i \tilde{C}_{pi}(T - T_{i0})}_{3} = 0$$

This equation has been divided into three parts. Which of these parts represent heat added to the system and which of these represent heat removed from the system? [25]

Heat Added (positive quantities) = 1, 2

Heat Removed (negative quantities) = 3, 1 + 16

3. For the reaction in question #1, the reactants enter the CSTR at 275 K and the CSTR is run adiabatically. If $F_{A0} = F_{B0} = 8 \text{ mol/s}$, $C_{PA} = C_{PB} = C_{PC} = C_{PD} = 400 \text{ J/(mol}^\circ\text{C)}$, and final temperature is 375 K, find the value of the heat change due to convection/flow. [25]

$$\Theta_A = \Theta_B = 1$$

$$\begin{aligned} -F_{A0} \sum_{i=1}^n \Theta_i C_{pi}(T - T_{i0}) &= -8 \frac{\text{mol}}{\text{s}} \left[\Theta_A C_{pA}(T - T_{i0}) + \Theta_B C_{pB}(T - T_{i0}) \right] \\ &= -8 \frac{\text{mol}}{\text{s}} \left[400 \frac{\text{J}}{\text{mol}^\circ\text{C}} (375 - 275) + 400 (375 - 275) \right] \\ &= \left[-640 \frac{\text{kJ}}{\text{s}} \right] \end{aligned}$$

4. What is the conversion for problem #3?

$$[-\Delta H_{Rx}^0(T_R) - \Delta \hat{C}_p(T - T_R)]F_{A0}X - 640 \frac{\text{kJ}}{\text{s}} = 0$$

$$X = \frac{640 \frac{\text{kJ}}{\text{s}}}{[+\Delta H_{Rx}(T_R) - \Delta \hat{C}_p(T - T_R)]F_{A0}} = \frac{640}{[+100 - \Delta C_p(375 - 275)] 8 \frac{\text{mol}}{\text{s}}}$$